

Non-Singular Vacuum Mechanics: The Logarithmic Density Ceiling, Acoustic Mach Boundaries, and Saturated Core Regularization in Geotopological Hydrodynamics

Taquan Abram

Professional Engineer, Flint, Michigan, USA

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ABSTRACT

We formalize the non-singular vacuum mechanism of Geotopological Hydrodynamics (GTH v12.0), in which the physical vacuum is modeled as a 5D complex-coherent superfluid substrate. Classical General Relativity (GR) predicts that catastrophic gravitational collapse leads to point-like geometric singularities ($r \rightarrow 0$, $R_{\mu\nu\alpha\beta} R^{\mu\nu\alpha\beta} \rightarrow \infty$) and the black hole information loss paradox. In GTH, density protection is established as an exact, non-perturbative property of the substrate. We demonstrate that the topological potential $V_{\text{top}}(\rho) = \lambda_{\text{top}} [-\ln(1 - 2\alpha\kappa\rho) + \frac{1}{2} \|d M^{(3)}\|^2]$ generates an insurmountable logarithmic energy barrier as local condensate density approaches the finite upper ceiling $\rho_{\text{max}} = 1 / (2\alpha\kappa)$. We prove that the barrier derivative $\partial V_{\text{top}} / \partial \rho$ diverges as $\rho \rightarrow \rho_{\text{max}}^-$, strictly forbidding infinite-density singularities in all finite-action states. We derive the exact minimum core radius inequality $R_c \geq (3 M_c / (4\pi \rho_{\text{max}}))^{1/3}$ for compact objects of core mass M_c , prove the monotonicity and convexity of the logarithmic density potential, and show how non-linear Carreau-Yasuda rheology replaces classical event horizons with regular, shear-thinning solitonic acoustic Mach surfaces where $|v| = c_s$. Finally, we detail the implementation of this density ceiling as a continuous interior barrier penalty $P_{\text{inad}}(\Theta)$ within grand MCMC likelihood inversions across galactic, cluster, and wave channels.

1. INTRODUCTION & THE SINGULARITY PROBLEM IN GENERAL RELATIVITY

The Penrose-Hawking singularity theorems establish that under standard energy conditions, classical General Relativity inevitably produces spacetime singularities in gravitational collapse. At these singular boundaries, curvature invariants such as the Kretschmann scalar $K_S = R_{\mu\nu\alpha\beta} R^{\mu\nu\alpha\beta}$ diverge, differential manifold structures break down, and physical determinism fails.

In Geotopological Hydrodynamics (GTH v12.0), the vacuum is modeled as a continuous, compressible 5D superfluid substrate (the Omicron Condensate) characterized by the irreducible constitutive state vector:

$$\Theta \equiv (M_{\text{UV}}, m_{\text{IR}}, \rho_0, K, G_{\text{shear}}, \tau_0, \eta_n)$$

Where $M_{\text{UV}} = 2.1570 \times 10^{-8}$ kg, $m_{\text{IR}} = 1.8184 \times 10^{-69}$ kg, $\rho_0 = 1.0100 \times 10^{-26}$ kg m⁻³, $K = 1.5150 \times 10^{-10}$ Pa, $G_{\text{shear}} = 8.0797 \times 10^{-11}$ Pa, $\tau_0 = 1.2500 \times 10^{-2}$ s, and $\eta_n = 1.1500 \times 10^{-12}$ Pa s. Longitudinal compression and transverse shear propagate at distinct material speeds:

$$c_s = \sqrt{\frac{K}{\rho_0}} = 1.2247 \times 10^8 \text{ m s}^{-1}, \quad c_{\text{substrate}} = \sqrt{\frac{G_{\text{shear}}}{\rho_0}} = 8.9443 \times 10^7 \text{ m s}^{-1}$$

Gravitational interaction emerges as the long-wavelength elastic response of this substrate, with the emergent weak-field coupling derived via zero-mode projection as:

$$G_{\text{model}} = \frac{3\pi\hbar c_s}{4M_{\text{UV}}^2} = 6.5406 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$$

2. MATHEMATICAL FORMULATION OF THE LOGARITHMIC DENSITY BARRIER

The macroscopic order parameter of the substrate is represented as a complex scalar field $\Psi(X) = \sqrt{\rho(X)} e^{i\phi(X)}$. The 5D GTH action functional S_{GTH} incorporates a topological potential density $V_{\text{top}}(\rho)$:

$$V_{\text{top}}(\rho) = \lambda_{\text{top}} \left[-\ln(1 - \Xi(\rho)) + \frac{1}{2} \|dM^{(3)}\|^2 \right], \quad \Xi(\rho) \equiv 2\alpha\kappa\rho < 1$$

THEOREM 1 (Monotonicity and Divergence of the Density Potential).

For any physical state with density $0 \leq \rho < \rho_{\text{max}} \equiv 1 / (2\alpha\kappa)$:

(i) The potential $V_{\text{top}}(\rho)$ is strictly monotonic and convex.

(ii) The barrier derivative $\partial V_{\text{top}} / \partial \rho$ diverges strictly as $\rho \rightarrow \rho_{\text{max}}^-$.

$$\frac{\partial V_{\text{top}}}{\partial \rho} = \frac{2\alpha\kappa\lambda_{\text{top}}}{1 - 2\alpha\kappa\rho} \longrightarrow +\infty \quad \text{as } \rho \rightarrow \rho_{\text{max}}^- \equiv \left(\frac{1}{2\alpha\kappa}\right)^-$$

Proof: Direct differentiation of $V_{\text{top}}(\rho)$ with respect to ρ yields $\partial V_{\text{top}} / \partial \rho = 2\alpha\kappa\lambda_{\text{top}} / (1 - 2\alpha\kappa\rho)$. Because λ_{top} , α , and κ are positive, $\partial V_{\text{top}} / \partial \rho > 0$ for all $\rho \in [0, \rho_{\text{max}})$. The second derivative $\partial^2 V_{\text{top}} / \partial \rho^2 = 4\alpha^2\kappa^2\lambda_{\text{top}} / (1 - 2\alpha\kappa\rho)^2 > 0$ proves strict convexity. Taking $\rho \rightarrow \rho_{\text{max}}^-$ yields $+\infty$; Q.E.D.

3. COMPACT CORE REGULARIZATION & MINIMUM CORE RADIUS INEQUALITY

THEOREM 2 (Minimum Core Radius Bound).

In any admissible configuration governed by S_{GTH} , the physical core radius R_c of a compact object of mass M_c satisfies the strict lower bound:

$$R_c \geq \left(\frac{3M_c}{4\pi\rho_{\text{max}}} \right)^{1/3} = \left(\frac{3\alpha\kappa M_c}{2\pi} \right)^{1/3} > 0$$

Proof: The average spatial mass density $\rho_{\text{c}}^{\text{■}}$ within core volume $V_c = (4/3)\pi R_c^3$ is $\rho_{\text{c}}^{\text{■}} = 3M_c / (4\pi R_c^3)$. Admissibility under Theorem 1 requires $\rho_{\text{c}}^{\text{■}} \leq \rho_{\text{max}} = 1/(2\alpha\kappa)$. Inverting this inequality gives $R_c^3 \geq 3M_c / (4\pi\rho_{\text{max}})$, yielding the core radius lower bound. Q.E.D.

4. ACOUSTIC MACH SURFACES & CARREAU-YASUDA RHEOLOGY

In GTH, classical black hole event horizons are replaced by acoustic Mach surfaces where inward substrate flow speed equals longitudinal sound speed $|v(r_h)| = c_s$. The effective radial acoustic metric takes Painlevé-Gullstrand form:

$$ds_{\text{eff}}^2 = -(c_s^2 - v^2)dt^2 - 2vdrdt + dr^2 + r^2d\Omega^2$$

Under extreme shear strain rates $\dot{\gamma}^{\text{■}}$ near the sonic boundary, non-linear Carreau-Yasuda operators quench dynamic viscosity ($\eta_{\text{eff}} \rightarrow 0$):

$$\tau_{\text{eff}}(\dot{\gamma}) = \tau_0 [1 + (\lambda_{\tau}\dot{\gamma})^{a_{\tau}}]^{\frac{n_{\tau}-1}{a_{\tau}}}, \quad \eta_{\text{eff}}(\dot{\gamma}) = \eta_n [1 + (\lambda_{\eta}\dot{\gamma})^{a_{\eta}}]^{\frac{n_{\eta}-1}{a_{\eta}}}$$

Under extreme collapse, the substrate dumps viscoelastic memory and vents excess inflow energy as outgoing sub-critical shear waves rather than forming singular coordinate states.

5. LOGARITHMIC INTERIOR BARRIER TELEMETRY IN MCMC INVERSIONS

To enforce $\rho < \rho_{\text{max}}$ during Bayesian parameter estimation across the 175 SPARC galaxies, cluster dynamics, and wave sectors, GTH applies a logarithmic interior-point penalty $P_{\text{inad}}(\Theta)$:

$$P_{\text{inad}}(\Theta) \equiv -\mu_{\text{barrier}} \sum_b \ln(1 - \Xi_b(\Theta))$$

$$-2\ln \mathcal{L}_{\text{joint}}(\Theta) = \Delta d^T C_{\text{joint}}^{-1} \Delta d + \ln \det C_{\text{joint}} + N_{\text{data}} \ln(2\pi) + 2P_{\text{inad}}(\Theta)$$

6. CONCLUSION & SERIES ROADMAP

Geotopological Hydrodynamics replaces classical GR point singularities with finite-density saturated cores of radius $R_c \geq (3 M_c / (4\pi \rho_{\text{max}}))^{1/3} > 0$, resolving the black hole singularity and information loss paradoxes.

Paper #	Monograph Title / Physical Focus	Key Formulation
Paper 1	Density Ceiling & Core Regularization	$V_{\text{top}}(\rho) \rightarrow +\infty, R_c \geq (3M_c / 4\pi \rho_{\text{max}})^{1/3}$
Paper 2	Scale Isolation & Chiral Orthogonality	$M_{\{G,\text{eff}\}}^2 = M_G^2 / (1 + Z(\text{grad } \phi))$
Paper 3	Beltrami Vortex Wakes & Disk Dynamics	$\beta(r) = \chi_n G_{\text{model}} \Sigma_b(r) e^{(-\text{Re}_G \text{TH})}$
Paper 4	Acoustic Horizons & Echo Comb	$\tau_{\text{echo}}(\omega) = 2 (d \Phi_{\text{sv}} / d \omega)$
Paper 5	Zero-Mode Gauss-Codazzi Projection	$G^{(4)}_{\{\mu \nu\}} + \Lambda_{\text{curv}} h_{\{\mu \nu\}} = \kappa_{\text{SI}} T^{\{\text{eff}\}}_{\{\mu \nu\}} + C_{\{\mu \nu\}}$